Machine Learning - Assignment 1

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Problem 1: Multi-output Linear Regression

1.1 Maximum Likelihood Estimation for W

The probabilistic model assumes that for a given input x_i, the response vector y_i is drawn from a multivariate normal distribution:

$$\mathbf{y}_i|x_i \sim \mathcal{N}(\mathbf{W}^\top \phi(x_i), \mathbf{\Sigma}) \quad \text{with} \quad \mathbf{\Sigma} = \sigma^2 I_2$$

The log-likelihood function for the entire dataset $D = \{(x_i, \mathbf{y}_i)\}_{i=1}^n$ is given by:

$$\begin{aligned}
\mathcal{L}(\mathbf{W}, \mathbf{\Sigma}; D) &= \log \prod_{i=1}^n P(\mathbf{y}_i|x_i; \mathbf{W}, \mathbf{\Sigma}) \\
&= \sum_{i=1}^n \log P(\mathbf{y}_i|x_i; \mathbf{W}, \mathbf{\Sigma}) \\
&= \sum_{i=1}^n \log \left[\left(\frac{1}{(2\pi)^k |\mathbf{\Sigma}|} \right)^{\frac{1}{2}} \exp\left(-\frac{1}{2} (\mathbf{y}_i - \mathbf{W}^\top \phi(x_i))^\top \mathbf{\Sigma}^{-1} (\mathbf{y}_i - \mathbf{W}^\top \phi(x_i)) \right) \right] \\
&= \sum_{i=1}^n \left(-\frac{k}{2} \log(2\pi) - \frac{1}{2} \log |\mathbf{\Sigma}| - \frac{1}{2} (\mathbf{y}_i - \mathbf{W}^\top \phi(x_i))^\top \mathbf{\Sigma}^{-1} (\mathbf{y}_i - \mathbf{W}^\top \phi(x_i)) \right)
\end{aligned}$$

To find the maximum likelihood estimate for $\mathbf{W}$, we need to maximize $\mathcal{L}(\mathbf{W}, \mathbf{\Sigma}|D)$ with respect to $\mathbf{W}$. This is equivalent to minimizing the negative log-likelihood, and since the terms not involving $\mathbf{W}$ are constant with respect to $\mathbf{W}$, this simplifies to only minimizing the sum of the quadratic forms:

$$\hat{\mathbf{W}}_{\text{MLE}} = \arg\min_{\mathbf{W}} \sum_{i=1}^n (\mathbf{y}_i - \mathbf{W}^\top \phi(x_i))^\top \mathbf{\Sigma}^{-1} (\mathbf{y}_i - \mathbf{W}^\top \phi(x_i))$$

Now let $\mathbf{w}_1$ and $\mathbf{w}_2$ be the first and second columns of $\mathbf{W}$, respectively. Due to the one-hot encoding of $\phi(x)$, the predicted mean $\hat{\mathbf{y}}_i = \mathbf{W}^\top \phi(x_i)$ simplifies to:

$$\hat{\mathbf{y}}_i = \begin{cases} \mathbf{W}^\top \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \mathbf{w}_1 & \text{if } x_i = 0 \\ \mathbf{W}^\top \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \mathbf{w}_2 & \text{if } x_i = 1 \end{cases}$$

Now let $D_0 = \{\mathbf{y}_i | x_i = 0\}$ and $D_1 = \{\mathbf{y}_i | x_i = 1\}$. The minimization problem decouples into two independent problems for $\mathbf{w}_1$ and $\mathbf{w}_2$:

$$\min_{\mathbf{w}_1} \sum_{\mathbf{y}_i \in D_0} (\mathbf{y}_i - \mathbf{w}_1)^\top \mathbf{\Sigma}^{-1} (\mathbf{y}_i - \mathbf{w}_1) \quad \text{and} \quad \min_{\mathbf{w}_2} \sum_{\mathbf{y}_i \in D_1} (\mathbf{y}_i - \mathbf{w}_2)^\top \mathbf{\Sigma}^{-1} (\mathbf{y}_i - \mathbf{w}_2)$$

The solution that minimizes the sum of squared Mahalanobis distances to a set of points is the sample mean of those points. Thus:

$$\hat{\mathbf{w}}_1 = \frac{1}{|D_0|} \sum_{\mathbf{y}_i \in D_0} \mathbf{y}_i \quad \text{and} \quad \hat{\mathbf{w}}_2 = \frac{1}{|D_1|} \sum_{\mathbf{y}_i \in D_1} \mathbf{y}_i$$

For our data:

$$D_0 = \left\{ \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \end{pmatrix} \right\}, \quad D_1 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

Calculating the means:

$$\hat{\mathbf{w}}_1 = \frac{1}{3} \begin{pmatrix} -1-1-2 \\ -1-2-1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -4 \\ -4 \end{pmatrix} = \begin{pmatrix} -4/3 \\ -4/3 \end{pmatrix}$$

$$\hat{\mathbf{w}}_2 = \frac{1}{3} \begin{pmatrix} 1+1+2 \\ 1+2+1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 4/3 \end{pmatrix}$$

The MLE for $\mathbf{W}$ is constructed by combining these column vectors:

$$\hat{\mathbf{W}}_{\text{MLE}} = \begin{pmatrix} | & | \\ \hat{\mathbf{w}}_1 & \hat{\mathbf{w}}_2 \\ | & | \end{pmatrix} = \begin{pmatrix} -4/3 & 4/3 \\ -4/3 & 4/3 \end{pmatrix}$$

1.2 Ordinary Least Squares Estimation for W

The OLS estimator minimizes the residual sum of squares (RSS):

$$\text{RSS}(\mathbf{W}; D) = \sum_{i=1}^{n} \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2^2 = \sum_{i=1}^{n} \|\mathbf{y}_i - \mathbf{W}^\top \phi(x_i)\|_2^2$$

Using the same decomposition based on the value of x_i:

$$\text{RSS}(\mathbf{W}; D) = \sum_{\mathbf{y}_i \in D_0} \|\mathbf{y}_i - \mathbf{w}_1\|^2 + \sum_{\mathbf{y}_i \in D_1} \|\mathbf{y}_i - \mathbf{w}_2\|^2$$

This objective function also decouples. To minimize the total sum, we can minimize each part independently. The vector $\mathbf{w}_1$ that minimizes the sum of squared Euclidean distances to the points in D_0 is the sample mean of D_0. Similarly, the optimal $\mathbf{w}_2$ is the sample mean of D_1.

$$\hat{\mathbf{w}}_1 = \frac{1}{|D_0|} \sum_{\mathbf{y}_i \in D_0} \mathbf{y}_i = \begin{pmatrix} -4/3 \\ -4/3 \end{pmatrix}$$

$$\hat{\mathbf{w}}_2 = \frac{1}{|D_1|} \sum_{\mathbf{y}_i \in D_1} \mathbf{y}_i = \begin{pmatrix} 4/3 \\ 4/3 \end{pmatrix}$$

Thus, the OLS estimate for $\mathbf{W}$ is:

$$\hat{\mathbf{W}}_{\text{OLS}} = \begin{pmatrix} | & | \\ \hat{\mathbf{w}}_1 & \hat{\mathbf{w}}_2 \\ | & | \end{pmatrix} = \begin{pmatrix} -4/3 & 4/3 \\ -4/3 & 4/3 \end{pmatrix} = \hat{\mathbf{W}}_{\text{MLE}}$$

According to the calculation results, the maximum likelihood estimator and the ordinary least squares estimator for the weight matrix $\mathbf{W}$ are the same in this problem. In fact, $\hat{\mathbf{W}}_{\text{MLE}} \equiv \hat{\mathbf{W}}_{\text{OLS}}$.

Problem 2: Linear Regression — MLE, MAP, and Bayesian Inference

2.1 Maximum Likelihood Estimation

The likelihood of a single data point y_i given x_i and a is:

$$p(y_i|x_i; a, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - ax_i)^2}{2\sigma^2}\right)$$

Since the data points are independent, the likelihood of all data points is the product of individual likelihoods:

$$L(a) = p(y_1, \ldots, y_n|x_1, \ldots, x_n; a, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - ax_i)^2}{2\sigma^2}\right)$$

To find the MLE for a, we minimize the negative log-likelihood function $\log L(a)$:

$$\text{NLL}(a) = -\log L(a) = \sum_{i=1}^{n} \left[\frac{1}{2}\log(2\pi\sigma^2) + \frac{(y_i - ax_i)^2}{2\sigma^2}\right]$$

To find the value of a that minimizes $\text{NLL}(a)$, we take the derivative with respect to a and set it to 0:

$$\frac{\partial \text{NLL}(a)}{\partial a} = \sum_{i=1}^{n} \frac{\partial}{\partial a}\left[\frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}(y_i - ax_i)^2\right] = 0$$

The first term is constant with respect to a, so its derivative is 0. For the second term, we use the chain rule:

$$\frac{\partial}{\partial a}\left(-\frac{1}{2\sigma^2}(y_i - ax_i)^2\right) = -\frac{1}{2\sigma^2} \cdot 2(y_i - ax_i) \cdot (-x_i) = \frac{1}{\sigma^2}(y_i - ax_i)x_i$$

Summing over all i:

$$\begin{aligned}
\frac{\partial \log L(a)}{\partial a} = \sum_{i=1}^{n} \frac{1}{\sigma^2}(y_i x_i - a x_i^2) &= 0 \\
\sum_{i=1}^{n}(y_i x_i - a x_i^2) &= 0 \\
\sum_{i=1}^{n} y_i x_i - a \sum_{i=1}^{n} x_i^2 &= 0
\end{aligned}$$

Solving for a:

$$\hat{a}_{\text{MLE}} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$

2.2 Maximum A Posterior Estimation

For MAP case, we want to find:

$$\hat{a}_{\text{MAP}} = \arg\max_a \left[\log p(y_1, \ldots, y_n | x_1, \ldots, x_n, a, \sigma^2) + \log p(a|\lambda)\right]$$

First, the log-likelihood with $\sigma = 1$:

$$\log p(y|x, a, \sigma^2)\big|_{\sigma=1} = \sum_{i=1}^{n} \left[-\frac{1}{2}\log(2\pi) - \frac{1}{2}(y_i - ax_i)^2\right]$$

Next, the prior distribution for a:

$$p(a|\lambda) = \frac{1}{\sqrt{2\pi\lambda^2}} \exp\left(-\frac{a^2}{2\lambda^2}\right)$$

The log-prior function:

$$\log p(a|\lambda) = -\frac{1}{2}\log(2\pi\lambda^2) - \frac{a^2}{2\lambda^2} = -\frac{1}{2}\log(2\pi) - \log(\lambda) - \frac{a^2}{2\lambda^2}$$

Now, we combine the log-likelihood and log-prior to form the function $f(a)$ to be maximized (or minimize the negative, whatever):

$$f(a) = \log p(y|x, a, \sigma^2) + \log p(a|\lambda)$$

To find $\hat{a}_{\text{MAP}}$, we take the derivative of $f(a)$ with respect to a and set it to zero:

$$\begin{aligned}\frac{\partial f(a)}{\partial a} &= \frac{\partial}{\partial a}\left(\sum_{i=1}^{n}\left[-\frac{1}{2}\log(2\pi) - \frac{1}{2}(y_i - ax_i)^2\right]\right) + \frac{\partial}{\partial a}\left(-\frac{1}{2}\log(2\pi) - \log(\lambda) - \frac{a^2}{2\lambda^2}\right) \\ &= \sum_{i=1}^{n}(y_i x_i - a x_i^2) - \frac{a}{\lambda^2} = 0\end{aligned}$$

Solving for a:

$$\hat{a}_{\text{MAP}} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2 + \frac{1}{\lambda^2}}$$

2.3 Qualitative Comparison

Condition	$p(a\|\lambda)$	$p(Y\|X, a)$	$\|\hat{a}_{\textbf{MLE}} - \hat{a}_{\textbf{MAP}}\|$
$\lambda \to \infty$	wider	same	decrease
$\lambda \to 0$	narrower	same	increase
$n \to \infty$ (fixed λ)	same	narrower	decrease

- **As $\lambda \to \infty$:** The prior $p(a|\lambda) \sim \mathcal{N}(0, \lambda^2)$ becomes infinitely wide, indicating a very weak prior belief. The likelihood remains unchanged as it does not depend on λ. As $\frac{1}{\lambda^2} \to 0$, $\hat{a}_{\text{MAP}}$ approaches $\hat{a}_{\text{MLE}}$, so the distance between them decreases.
- **As $\lambda \to 0$:** The prior $p(a|\lambda) \sim \mathcal{N}(0, \lambda^2)$ becomes infinitely narrow, concentrating all probability mass at $a = 0$. The likelihood remains unchanged. As $\frac{1}{\lambda^2} \to \infty$, the denominator of $\hat{a}_{\text{MAP}}$ becomes very large, forcing $\hat{a}_{\text{MAP}} \to 0$. Since $\hat{a}_{\text{MLE}}$ is generally not zero, the distance between them increases.
- **As $n \to \infty$ (fixed λ):** The prior $p(a|\lambda)$ depends only on λ, so it remains unchanged. With more data, the likelihood function becomes more concentrated (narrower) around the true parameter value, as the data provides more information. As $n \to \infty$, $\sum_{i=1}^{n} x_i^2$ typically grows large, making $\frac{1}{\lambda^2}$ negligible compared to $\sum_{i=1}^{n} X_i^2$. Thus, $\hat{a}_{\text{MAP}}$ approaches $\hat{a}_{\text{MLE}}$, and the distance between them decreases. This illustrates that with sufficient data, the likelihood dominates the prior in Bayesian inference.

Problem 3: Solving LASSO

3.1 Derivation of Equivalence

The MAP estimator for β is defined as:

$$\hat{\beta}_{\text{MAP}} = \arg\max_{\beta} \left[p(Y|X,\beta) + p(\beta|\lambda)\right]$$

Which is equivalent to minimizing the negative log-likelihood and prior:

$$\hat{\beta}_{\text{MAP}} = \arg\min_{\beta} \left[-\log p(Y|X,\beta) - \log p(\beta|\lambda)\right]$$

In which:

$$\begin{aligned}
p(Y|X,\beta) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - X_i\beta)^2}{2\sigma^2}\right) \\
\log p(Y|X,\beta) &= \sum_{i=1}^{n} \left[\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(Y_i - X_i\beta)^2}{2\sigma^2}\right] \\
&= \sum_{i=1}^{n} \left[-\frac{1}{2}\log 2\pi - \log\sigma\right] - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - X_i\beta)^2 \\
&= \sum_{i=1}^{n} \left[-\frac{1}{2}\log 2\pi - \log\sigma\right] - \frac{1}{2\sigma^2}\|Y - X\beta\|_2^2
\end{aligned}$$

and,

$$\begin{aligned}
p(\beta|\lambda) &= \prod_{j=1}^{p} \frac{\lambda}{2} \exp(-\lambda|\beta_j|) \\
\log p(\beta|\lambda) &= \sum_{j=1}^{p} \left(\log\left(\frac{\lambda}{2}\right) - \lambda|\beta_j|\right) \\
&= p\log\left(\frac{\lambda}{2}\right) - \lambda\sum_{j=1}^{p}|\beta_j| \\
&= p\log\left(\frac{\lambda}{2}\right) - \lambda\|\beta\|_1
\end{aligned}$$

Removing the constant terms that do not depend on β:

$$\begin{aligned}
\hat{\beta}_{\text{MAP}} &= \arg\min_{\beta} \left(\frac{1}{2\sigma^2}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1\right) \\
&= \arg\min_{\beta} \left(\|Y - X\beta\|_2^2 + \lambda'\|\beta\|_1\right)
\end{aligned}$$

Where $\lambda' = \frac{\lambda}{\sigma^2}$ is regularized λ. Thus, the MAP estimation is proved to be equivalent to solving a LASSO optimization problem.

3.2 Subgradient Update Rule for LASSO

Given the LASSO problem:

$$\min_{\beta} L(\beta) = \frac{1}{2n}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1$$

The subgradient can be split into two parts:

$$\nabla L(\beta) = \nabla\left(\frac{1}{2n}\|Y - X\beta\|_2^2\right) + \nabla\left(\lambda\|\beta\|_1\right)$$

For each part:

$$\begin{aligned}
\nabla\left(\frac{1}{2n}\|Y - X\beta\|_2^2\right) &= \nabla\left(\frac{1}{2n}(Y - X\beta)^T(Y - X\beta)\right) \\
&= \frac{1}{2n}\nabla\left(Y^TY - 2Y^TX\beta + \beta^TX^TX\beta\right) \\
&= \frac{1}{2n}(-2X^TY + 2X^TX\beta) \\
&= \frac{1}{n}X^T(X\beta - Y)
\end{aligned}$$

and,

$$\nabla\left(\lambda\|\beta\|_1\right) = \lambda\nabla\left(\sum_{j=1}^{p}|\beta_j|\right) = \lambda\begin{pmatrix}\operatorname{sgn}(\beta_1)\\ \operatorname{sgn}(\beta_2)\\ \vdots \\ \operatorname{sgn}(\beta_p)\end{pmatrix} = \lambda\ \operatorname{sgn}(\beta),\ \operatorname{sgn}(\beta_j) = \begin{cases}1 & \beta_j > 0\\ -1 & \beta_j < 0\\ s_j \in [-1,1] & \beta_j = 0\end{cases}$$

For the purpose of subgradient descent, a common choice for sgn(0) is 0.

Thus, the subgradient update rule for LASSO is:

$$\begin{aligned}
\beta^{(k+1)} &= \beta^{(k)} - \alpha_k \cdot g \\
&= \beta^{(k)} - \alpha_k\left[\frac{1}{n}X^T(X\beta^{(k)} - Y) + \lambda\ \operatorname{sgn}(\beta^{(k)})\right]
\end{aligned}$$

3.3 Coordinate Descent for LASSO

Coordinate descent optimizes one component β_j of β at a time, holding all other components fixed.

Now let $\mathbf{x}_j$ denote the j-th column of matrix X, we can re-write $X\beta$ as $\mathbf{x}_j\beta_j + \sum_{k\neq j}\mathbf{x}_k\beta_k$. Let $Y^{(j)} = Y - \sum_{k\neq j}\mathbf{x}_k\beta_k$ be the partial residual, which includes the effect of all β_k where $k \neq j$. Thus the objective function with respect to β_j (keeping all other β_k fixed) becomes:

$$L(\beta_j) = \frac{1}{2n}\|Y^{(j)} - \mathbf{x}_j\beta_j\|_2^2 + \lambda|\beta_j| + C$$

where C represents terms independent of β_j.

Expanding the squared Euclidean norm:

$$L(\beta_j) = \frac{1}{2n}\left((Y^{(j)})^TY^{(j)} - 2(Y^{(j)})^T\mathbf{x}_j\beta_j + (\mathbf{x}_j^T\mathbf{x}_j)\beta_j^2\right) + \lambda|\beta_j| + C$$

Let $A = \frac{1}{n}\mathbf{x}_j^T\mathbf{x}_j = \frac{1}{n}\|\mathbf{x}_j\|_2^2$ and $B = \frac{1}{n}(Y^{(j)})^T\mathbf{x}_j$. The objective function (ignoring constants) simplifies to:

$$L(\beta_j) = \frac{1}{2}A\beta_j^2 - B\beta_j + \lambda|\beta_j|$$

To find the optimal β_j, we take the subgradient with respect to β_j and set it to zero:

$$A\beta_j - B + \lambda \cdot \text{sgn}(\beta_j) = 0$$

where $\text{sgn}(\beta_j)$ is the subgradient of $|\beta_j|$.

There are three cases:

1. If $\beta_j > 0$: $A\beta_j - B + \lambda = 0 \implies \beta_j = \frac{B-\lambda}{A}$. This solution is valid only if $\frac{B-\lambda}{A} > 0$, which implies $B > \lambda$.
2. If $\beta_j < 0$: $A\beta_j - B - \lambda = 0 \implies \beta_j = \frac{B+\lambda}{A}$. This solution is valid only if $\frac{B+\lambda}{A} < 0$, which implies $B < -\lambda$.
3. If $\beta_j = 0$: The subgradient equation becomes $-B + \lambda \cdot s = 0$ for some $s \in [-1, 1]$. This implies $B = \lambda s$, which means $|B| \leq \lambda$.

Combining these cases, the update rule for β_j is given by the soft-thresholding:

$$\beta_j = \textit{soft-thresholding}(\frac{B}{A}, \frac{\lambda}{A}) = \text{sgn}(\frac{B}{A}) \max\left(0, |\frac{B}{A}| - \frac{\lambda}{A}\right)$$

Substituting back $A = \frac{1}{n}\|\mathbf{x}_j\|_2^2$ and $B = \frac{1}{n}(Y^{(j)})^T\mathbf{x}_j$:

$$\beta_j = \text{sgn}\left(\frac{(Y^{(j)})^T\mathbf{x}_j}{\|\mathbf{x}_j\|_2^2}\right) \max\left(0, \left|\frac{(Y^{(j)})^T\mathbf{x}_j}{\|\mathbf{x}_j\|_2^2}\right| - \frac{\lambda}{\frac{1}{n}\|\mathbf{x}_j\|_2^2}\right)$$

This can be simplified to:

$$\beta_j = \frac{\text{sgn}((Y^{(j)})^T\mathbf{x}_j)}{\|\mathbf{x}_j\|_2^2} \max\left(0, |(Y^{(j)})^T\mathbf{x}_j| - n\lambda\right)$$

Let $C_j = (Y^{(j)})^T\mathbf{x}_j = (Y - \sum_{k\neq j}\mathbf{x}_k\beta_k)^T\mathbf{x}_j$. Then the update rule is:

$$\beta_j = \frac{\text{sgn}(C_j)}{\|\mathbf{x}_j\|_2^2} \max\left(0, |C_j| - n\lambda\right)$$

3.4 Influence of λ

The influence of λ on the solution sparsity can be rigorously explained from the perspective of the cumulative distribution function (CDF).

Since the Laplace distribution is symmetric around zero, for $z \geq 0$, the PDF of $Z = |\beta_j|$ can be obtained by summing the probabilities for $\beta_j = z$ and $\beta_j = -z$:

$$\begin{aligned} p_{|\beta_j|}(z|\lambda) &= p(\beta_j = z|\lambda) + p(\beta_j = -z|\lambda) \\ &= \frac{\lambda}{2}e^{-\lambda z} + \frac{\lambda}{2}e^{-\lambda(-z)} = \lambda e^{-\lambda z}, \quad \text{for } z \geq 0 \end{aligned}$$

the CDF of $Z = |\beta_j|$ is given by:

$$\begin{aligned} CDF_{|\beta_j|}(z|\lambda) = P(|\beta_j| \leq z|\lambda) &= \int_0^z p_{|\beta_j|}(t|\lambda)dt = \int_0^z \lambda e^{-\lambda t}dt \\ &= \left[-e^{-\lambda t}\right]_0^z = (-e^{-\lambda z}) - (-e^0) = 1 - e^{-\lambda z} \end{aligned}$$

For $z < 0$, $CDF_{|\beta_j|}(z|\lambda) = 0$.

the partial derivative of $CDF_{|\beta_j|}(z|\lambda)$ with respect to λ is:

$$\begin{aligned}\frac{\partial}{\partial\lambda}CDF_{|\beta_j|}(z|\lambda) &= \frac{\partial}{\partial\lambda}(1-e^{-\lambda z}) \\ &= ze^{-\lambda z} \geq 0\end{aligned}$$

Which indicates that as λ increases, the probability that $|\beta_j|$ falls within a small interval $[0, z]$ (i.e., close to zero) increases. This implies that the probability mass of $|\beta_j|$ is shifted towards zero, making the distribution of $|\beta_j|$ more concentrated around zero.

Thus, a larger λ imposes a stronger prior belief that the coefficients β_j should be close to zero, thus leading to a more **sparse** solution in case of soft-thresholding.

From a slightly more emotional perspective, a larger λ would make it easier for B to fall within $|B| < \lambda$ (case 3), leading to more β_j being zero, and thus also leading to a sparse solution.

Problem 4: Shrinkage in Linear Regression

Let's first derive the functional form of $\hat{w}_k$ for each estimation method:

- **Ordinary Least Squares (OLS):**
  The OLS estimator minimizes $||y - Xw||_2^2$. The normal equation is $X^T X \hat{w}_{\text{OLS}} = X^T y$. Given $X^T X = I$, we have:

  $$\hat{w}_{\text{OLS}} = X^T y$$

  For the k-th coefficient[1]:

  $$\hat{w}_{\text{OLS},k} = (X^T y)_k = \mathbf{x}_k^T y$$

  Since $c_k = 2y^T \mathbf{x}_k$, we have $\mathbf{x}_k^T y = c_k/2$. Thus, the OLS estimator for the k-th coefficient is:

  $$\hat{w}_{\text{OLS},k} = \frac{c_k}{2}$$

  This is a linear function of c_k passing through the origin with a slope of 1/2.

- **Ridge Regression:**
  The Ridge Regression estimator minimizes $||y - Xw||_2^2 + \lambda_2 ||w||_2^2$. The solution is:

  $$\hat{w}_{\text{Ridge}} = (X^T X + \lambda_2 I)^{-1} X^T y$$

  Given $X^T X = I$:

  $$\hat{w}_{\text{Ridge}} = (I + \lambda_2 I)^{-1} X^T y = ((1 + \lambda_2) I)^{-1} X^T y = \frac{1}{1 + \lambda_2} X^T y$$

  For the k-th coefficient:

  $$\hat{w}_{\text{Ridge},k} = \frac{1}{1 + \lambda_2} (X^T y)_k = \frac{1}{1 + \lambda_2} \mathbf{x}_k^T y$$

  Substituting $\mathbf{x}_k^T y = c_k/2$:

  $$\hat{w}_{\text{Ridge},k} = \frac{1}{1 + \lambda_2} \frac{c_k}{2}$$

  This is also a linear function of c_k passing through the origin, but with a slope of $\frac{1}{2(1+\lambda_2)}$. Since $\lambda_2 \geq 0$, this slope is less than or equal to 1/2, indicating shrinkage towards zero compared to OLS.

- **LASSO Regression:**
  The LASSO Regression estimator minimizes $||y - Xw||_2^2 + \lambda_1 ||w||_1$. In the orthonormal design matrix setting, the solution for the k-th coefficient is given by the soft-thresholding:

  $$\begin{aligned} \hat{w}_{\text{LASSO},k} &= \textit{soft-thresholding}(\mathbf{x}_k^T y, \lambda_1/2) \\ &= \text{sign}(\mathbf{x}_k^T y) \cdot \max(0, |\mathbf{x}_k^T y| - \lambda_1/2) \end{aligned}$$

[1] For the sake of simplicity and a easier life, here we use a slightly different notation from the original problem: let $\mathbf{X}$ denotes the matrix, $\mathbf{X}_i$ (uppercase) denotes its i-th row, and $\mathbf{x}_j$ (lowercase) denotes its j-th column. This notation aligns with the modern linear algebra concept of "viewing a matrix as a sequence of column vectors."

Substituting $\mathbf{x}_k^T y = c_k/2$ and $\tau = \lambda_1/2$:

$$\hat{w}_{\text{LASSO},k} = \text{sign}\left(\frac{c_k}{2}\right) \cdot \max\left(0, \left|\frac{c_k}{2}\right| - \frac{\lambda_1}{2}\right) = \begin{cases} \frac{c_k}{2} - \frac{\lambda_1}{2} & \text{if } c_k > \lambda_1 \\ \frac{c_k}{2} + \frac{\lambda_1}{2} & \text{if } c_k < -\lambda_1 \\ 0 & \text{if } |c_k| \leq \lambda_1 \end{cases}$$

LASSO exhibits a "sparse" behavior, setting coefficients to zero for small values of c_k, and then linearly shrinking the non-zero coefficients towards zero.

4.1 Identifying curves

Based on the derived functional forms and observing the typical shapes of these estimators:

- **Curve (1) (Solid): OLS**
  Curve (1) is a straight line passing through the origin. When $c_k = 2$, $\hat{w}_k = 1$, this perfectly matches the OLS functional form $\hat{w}_{\text{OLS},k} = c_k/2$, which has a slope of 1/2.
- **Curve (2) (Dotted): Ridge Regression**
  Curve (2) is also a straight line passing through the origin, but its slope is less steep than Curve (1). For example, when $c_k = 2$, $\hat{w}_k = 0.5$ (Coincident with Curve (3)), which is less than 1 (the OLS value). This behavior is consistent with Ridge Regression, which shrinks coefficients towards zero by having a slope $\frac{1}{2(1+\lambda_2)} < 1/2$ (for $\lambda_2 > 0$).
- **Curve (3) (Dashed): LASSO Regression**
  Curve (3) shows a distinct behavior where $\hat{w}_k$ is exactly zero for a range of c_k values around zero (from $c_k = -1$ to $c_k = 1$). Outside this range, it increases linearly with a slope similar to OLS but shifted towards zero.

4.2 Determining λ_1 and λ_2

Based on the formula $\hat{w}_{\text{Ridge},k} = \frac{1}{1+\lambda_2}\frac{c_k}{2}$ derived above, λ_2 can be calculated as:

$$\lambda_2 = \frac{c_k}{2\hat{w}_{\text{Ridge},k}} - 1$$

when $c_k = 2$, $\hat{w}_k = 0.5$. Substituting these values into the Ridge formula, we get $\lambda_2 = 1$.

From the LASSO functional form, $\hat{w}_{\text{LASSO},k} = 0$ when $|c_k| \leq \lambda_1$. Observing Curve (3) (LASSO), the estimated coefficient $\hat{w}_k$ is zero for c_k values between -1 and 1. Thus, the threshold for non-zero coefficients is $\lambda_1 = 1$.

Problem 5: Naive Bayes Classifier and MLE

5.1 Model Assumptions

The density $p(x|y = k)$ can factorize as $\prod_{j=1}^{d} N(x_j|\mu_{k,j}, \sigma_{k,j}^2)$ due to the Naive Bayes assumption of conditional independence. Which states that, given the class $y = k$, the features $x_1, \ldots, x_d$ are conditionally independent. Thus, the joint probability of the features given the class can be written as the product of the individual feature probabilities given the class:

$$p(x|y = k) = p(x_1, \ldots, x_d|y = k) = \prod_{j=1}^{d} p(x_j|y = k) = \prod_{j=1}^{d} N(x_j|\mu_{k,j}, \sigma_{k,j}^2)$$

5.2 MLE Derivation

Now let $\Theta = \{\pi_k, \mu_{k,j}, \sigma_{k,j}^2\}$ be the set of all model parameters. The likelihood for all N i.i.d. samples is given by:

$$\begin{aligned}
L(\Theta) = \prod_{i=1}^{N} p(x_i, y_i|\Theta) &= \prod_{i=1}^{N} p(x_i|y_i, \Theta)P(y_i|\Theta) \\
&= \prod_{i=1}^{N} \left(\pi_{y_i} \prod_{j=1}^{d} N(x_{i,j}|\mu_{y_i,j}, \sigma_{y_i,j}^2) \right) \\
&= \left(\prod_{i=1}^{N} \pi_{y_i} \right) \left(\prod_{i=1}^{N} \prod_{j=1}^{d} N(x_{i,j}|\mu_{y_i,j}, \sigma_{y_i,j}^2) \right)
\end{aligned}$$

Thus, the log-likelihood is given by:

$$\begin{aligned}
\log L(\Theta) &= \sum_{i=1}^{N} \log \pi_{y_i} + \sum_{i=1}^{N} \sum_{j=1}^{d} \log N(x_{i,j}|\mu_{y_i,j}, \sigma_{y_i,j}^2) \\
&= \sum_{i=1}^{N} \log \pi_{y_i} + \sum_{i=1}^{N} \sum_{j=1}^{d} \left(-\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma_{y_i,j}^2) - \frac{(x_{i,j} - \mu_{y_i,j})^2}{2\sigma_{y_i,j}^2} \right)
\end{aligned}$$

To derive π_k, we need to maximize $\log L(\Theta)$ with respect to π_k subject to the constraint $\sum_{k=1}^{K} \pi_k = 1$. We use a Lagrange multiplier λ:

$$\begin{aligned}
L'(\Theta, \lambda) &= \sum_{i=1}^{N} \log \pi_{y_i} + \lambda \left(\sum_{k=1}^{K} \pi_k - 1 \right) \\
&= \sum_{k=1}^{K} N_k \log \pi_k + \lambda \left(\sum_{k=1}^{K} \pi_k - 1 \right)
\end{aligned}$$

Where N_k is the number of samples belong to class k. Taking the derivative with respect to π_k and setting it to zero:

$$\frac{\partial L'}{\partial \pi_k} = \frac{N_k}{\pi_k} + \lambda = 0 \implies \pi_k = -\frac{N_k}{\lambda}$$

and,

$$\frac{\partial L'}{\partial \lambda} = \sum_{k=1}^{K} \pi_k - 1 = -\frac{\sum_{k=1}^{K} N_k}{\lambda} - 1 = 0 \implies \lambda = -\sum_{k=1}^{K} N_k = -N$$

Thus,

$$\pi_k = \frac{N_k}{N}$$

To derive $\mu_{k,j}$ and $\sigma^2_{k,j}$, we first need to select the terms depending on which from $\log L(\Theta)$, that is:

$$\sum_{y_i=k} \left(-\frac{1}{2} \log(\sigma^2_{k,j}) - \frac{(x_{i,j} - \mu_{k,j})^2}{2\sigma^2_{k,j}} \right)$$

Take the derivative with respect to $\mu_{k,j}$ and set it to zero:

$$\begin{aligned}
\frac{\partial \log L}{\partial \mu_{k,j}} = \sum_{y_i=k} \left(-\frac{1}{2\sigma^2_{k,j}} \cdot 2(x_{i,j} - \mu_{k,j}) \cdot (-1) \right) &= \sum_{y_i=k} \frac{x_{i,j} - \mu_{k,j}}{\sigma^2_{k,j}} = 0 \\
\sum_{y_i=k} x_{i,j} - N_k \mu_{k,j} &= 0 \\
\implies \hat{\mu_{k,j}} &= \frac{1}{N_k} \sum_{y_i=k} x_{i,j}
\end{aligned}$$

Do the same thing for $\sigma^2_{k,j}$:

$$\begin{aligned}
\frac{\partial \log L}{\partial \sigma^2_{k,j}} = \sum_{y_i=k} \left(-\frac{1}{2\sigma^2_{k,j}} + \frac{(x_{i,j} - \mu_{k,j})^2}{2(\sigma^2_{k,j})^2} \right) &= 0 \\
\sum_{y_i=k} \left(-\sigma^2_{k,j} + (x_{i,j} - \mu_{k,j})^2 \right) &= 0 \quad \text{(Multiply by } 2(\sigma^2_{k,j})^2\text{)} \\
-N_k \sigma^2_{k,j} + \sum_{y_i=k} (x_{i,j} - \mu_{k,j})^2 &= 0 \\
\implies \hat{\sigma}^2_{k,j} &= \frac{1}{N_k} \sum_{y_i=k} (x_{i,j} - \hat{\mu}_{k,j})^2
\end{aligned}$$

5.3 Modified Assumption — Shared Variances

The likelihood changes as follows:

$$L(\Theta) = \prod_{i=1}^{N} \left(\pi_{y_i} \prod_{j=1}^{d} N(x_{i,j} | \mu_{y_i,j}, \sigma^2_j) \right)$$

The log-likelihood thus becomes:

$$\log L(\Theta) = \sum_{i=1}^{N} \log \pi_{y_i} + \sum_{i=1}^{N} \sum_{j=1}^{d} \left(-\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2_j) - \frac{(x_{i,j} - \mu_{y_i,j})^2}{2\sigma^2_j} \right)$$

The only difference is that σ^2_j no longer has a subscript k (or, y_i).

Since the terms involving π_k are independent of the variance assumption, the MLE for π_k remains unchanged:

$$\hat{\pi}_k = \frac{N_k}{N}$$

The $\mu_{k,j}$ also remains unchanged:

$$\hat{\mu}_{k,j} = \frac{1}{N_k} \sum_{y_i=k} x_{i,j}$$

The only difference is the σ_j^2. Now, σ_j^2 is shared across all classes. The terms in $\log L(\Theta)$ that depend on σ_j^2 are:

$$\sum_{i=1}^{N} \left(-\frac{1}{2} \log(\sigma_j^2) - \frac{(x_{i,j} - \mu_{y_i,j})^2}{2\sigma_j^2} \right)$$

Which can be re-write by grouping terms by class:

$$\begin{aligned}
\sum_{k=1}^{K} \sum_{y_i=k} \left(-\frac{1}{2} \log(\sigma_j^2) - \frac{(x_{i,j} - \mu_{k,j})^2}{2\sigma_j^2} \right) &= \sum_{k=1}^{K} \left(N_k \left(-\frac{1}{2} \log(\sigma_j^2) \right) - \sum_{y_i=k} \frac{(x_{i,j} - \mu_{k,j})^2}{2\sigma_j^2} \right) \\
&= -\frac{N}{2} \log(\sigma_j^2) - \sum_{k=1}^{K} \sum_{y_i=k} \frac{(x_{i,j} - \mu_{k,j})^2}{2\sigma_j^2}
\end{aligned}$$

Thus the derivative becomes:

$$\begin{aligned}
\frac{\partial \log L}{\partial \sigma_j^2} &= -\frac{N}{2\sigma_j^2} + \sum_{k=1}^{K} \sum_{y_i=k} \frac{(x_{i,j} - \mu_{k,j})^2}{2(\sigma_j^2)^2} = 0 \\
\implies \hat{\sigma}_j^2 &= \frac{1}{N} \sum_{k=1}^{K} \sum_{y_i=k} (x_{i,j} - \hat{\mu}_{k,j})^2 = \frac{1}{N} \sum_{i=1}^{N} (x_{i,j} - \hat{\mu}_{y_i,j})^2
\end{aligned}$$

5.4 Decision Boundary

The decision boundary between two classes, let's say, class k and class l, is determined by the points x where $P(y = k|x) = P(y = l|x)$. This is equivalent to $\log P(y = k|x) = \log P(y = l|x)$, or $\log \frac{P(y=k|x)}{P(y=l|x)} = 0$.

Using Bayes' theorem $P(y = k|x) = \frac{p(x|y=k)P(y=k)}{p(x)}$, the decision boundary is given by:

$$\begin{aligned}
\log P(y = k) + \log p(x|y = k) &= \log P(y = l) + \log p(x|y = l) \\
\log \pi_k + \sum_{j=1}^{d} \log N(x_j|\mu_{k,j}, \sigma_{k,j}^2) &= \log \pi_l + \sum_{j=1}^{d} \log N(x_j|\mu_{l,j}, \sigma_{l,j}^2) \\
\log \frac{\pi_k}{\pi_l} + \sum_{j=1}^{d} \left(-\frac{1}{2} \log(\sigma_{k,j}^2) - \frac{(x_j - \mu_{k,j})^2}{2\sigma_{k,j}^2} \right) &= \sum_{j=1}^{d} \left(-\frac{1}{2} \log(\sigma_{l,j}^2) - \frac{(x_j - \mu_{l,j})^2}{2\sigma_{l,j}^2} \right)
\end{aligned}$$

That is,

$$\log \frac{\pi_k}{\pi_l} + \sum_{j=1}^{d} \left(\frac{1}{2} \log \frac{\sigma_{l,j}^2}{\sigma_{k,j}^2} - \frac{(x_j - \mu_{k,j})^2}{2\sigma_{k,j}^2} + \frac{(x_j - \mu_{l,j})^2}{2\sigma_{l,j}^2} \right) = 0$$

Which indicates that the decision boundary will be a quadratic function of x, means that the boundaries are generally curved, e.g., ellipse, hyperbola, etc.

For **shared variances**, the equation becomes:

$$\log \frac{\pi_k}{\pi_l} + \sum_{j=1}^{d} \left(\frac{1}{2} \log \frac{\sigma_{l,j}^2}{\sigma_{k,j}^2} - \frac{(x_j - \mu_{k,j})^2}{2\sigma_{k,j}^2} + \frac{(x_j - \mu_{l,j})^2}{2\sigma_{l,j}^2} \right) = 0$$

$$\log \frac{\pi_k}{\pi_l} + \sum_{j=1}^{d} \left(-\frac{(x_j - \mu_{k,j})^2}{2\sigma_j^2} + \frac{(x_j - \mu_{l,j})^2}{2\sigma_j^2} \right) = 0$$

$$\log \frac{\pi_k}{\pi_l} + \sum_{j=1}^{d} \frac{1}{2\sigma_j^2} \left(-(x_j^2 - 2x_j\mu_{k,j} + \mu_{k,j}^2) + (x_j^2 - 2x_j\mu_{l,j} + \mu_{l,j}^2) \right) = 0$$

$$\log \frac{\pi_k}{\pi_l} + \sum_{j=1}^{d} \frac{1}{2\sigma_j^2} \left(2x_j(\mu_{k,j} - \mu_{l,j}) + (\mu_{l,j}^2 - \mu_{k,j}^2) \right) = 0$$

Which indicates that the decision boundary will be a linear function of x, means the boundaries becomes hyperplanes.